Introduction to Machine Learning

Sample Study Notes for Week 3: Embeddings, Semantic Search & RAG

This deliberately text-based PDF is supplied as the source document for the retrieval assignment and mini project. It contains seven numbered content pages so source references can be checked directly.

Topics: machine learning, supervised and unsupervised learning, model evaluation, overfitting, embeddings, similarity search, retrieval-augmented generation, citations, and abstention.

Introduction to Machine Learning

Learning objectives

These notes introduce machine learning, common learning settings, model evaluation, text embeddings, and retrieval-augmented generation. The examples are designed for a small document-retrieval exercise.

What is machine learning?

Machine learning is a field of computing in which a system learns patterns from experience or data and uses those patterns to make predictions or decisions, rather than being explicitly programmed with a separate rule for every case.

Typical workflow

A simple workflow is to define the task, collect and inspect data, prepare useful features, train a model, evaluate it on held-out data, and monitor its performance after deployment. Data quality and a clearly defined target matter throughout the workflow.

Example

A spam classifier can learn patterns from labelled messages and estimate whether a new message is spam. The training examples help the model learn a relationship between message features and labels.

Supervised Learning

Definition

Supervised learning trains a model on labelled examples. Each example contains input features and a known target, so the model can learn a mapping from inputs to target values.

Classification

Classification predicts a category or class. Examples include identifying an email as spam or not spam, recognising a type of flower, and predicting whether a transaction is fraudulent.

Regression

Regression predicts a numeric value. Examples include estimating a house price, forecasting daily demand, or predicting delivery time.

Training signal

During training, the model compares its predictions with the known targets and adjusts its parameters to reduce a loss function. The goal is to generalise to new examples rather than simply memorise training data.

Unsupervised Learning

Definition

Unsupervised learning looks for structure in data without target labels supplied for each example.

Clustering

Clustering groups similar items together. It can be used to segment customers by behaviour, group similar documents, or explore patterns in measurements.

Dimensionality reduction

Dimensionality reduction represents data with fewer variables while preserving useful structure. It can support visualisation, compression, or preprocessing.

Important caution

A cluster is a pattern found by an algorithm, not automatically a meaningful real-world category. People should examine the data and validate whether the grouping is useful for the intended task.

Training, Validation, and Overfitting

Data splits

The training set is used to fit model parameters. The validation set helps compare models and tune hyperparameters. The test set is reserved for a final, relatively unbiased evaluation after decisions are complete.

Overfitting

Overfitting occurs when a model learns details or noise specific to the training data and performs poorly on unseen data. A large gap between training performance and validation performance can be a warning sign.

Ways to reduce overfitting

Useful approaches include collecting more representative data, simplifying the model, regularization, data augmentation when appropriate, and early stopping based on validation performance.

Data leakage

Data leakage occurs when information that would not legitimately be available at prediction time influences training or evaluation. It can produce overly optimistic results. The test set should not be used repeatedly to tune a model.

Model Evaluation Metrics

Confusion matrix

For binary classification, a confusion matrix counts true positives, true negatives, false positives, and false negatives. These counts help explain the kinds of mistakes a classifier makes.

Accuracy

Accuracy is the fraction of predictions that are correct: (true positives + true negatives) divided by all predictions. Accuracy can be misleading when one class is much more common than another.

Precision

Precision is true positives divided by predicted positives. It answers: among the items predicted as positive, what fraction were truly positive?

Recall

Recall is true positives divided by actual positives. It answers: among all actual positive items, what fraction did the model find?

F1 score

The F1 score is the harmonic mean of precision and recall. It is useful when both false positives and false negatives matter. Metric choice should reflect the practical cost of different errors.

Text Embeddings and Similarity Search

What is an embedding?

A text embedding represents text as a numeric vector. A good semantic embedding places texts with related meanings near one another in vector space, even if they use different words.

Similarity

Cosine similarity compares the angle between two vectors. Larger cosine similarity usually indicates that the vectors point in a more similar direction. Similarity is a ranking signal, not proof that a passage entails a claim.

Sparse and dense vectors

TF-IDF creates sparse vectors based on word importance and is a useful lightweight baseline. A sentence-transformer model such as all-MiniLM-L6-v2 creates dense contextual embeddings and can better capture semantic similarity, but its model files may need to be downloaded on first use.

Retrieval workflow

A retrieval system splits a document into chunks, embeds each chunk, embeds the question, calculates similarity, and returns the highest-ranking chunks. Smaller chunks offer more focused evidence; larger chunks preserve more context.

RAG, Sources, and Abstention

Retrieval-augmented generation

Retrieval-augmented generation (RAG) retrieves relevant passages from a source collection and provides those passages as context to a language model that generates an answer. Retrieval reduces the need to rely only on a model's parametric memory.

Chunks and page metadata

A chunk is a manageable passage of a document. Each chunk should keep metadata such as source filename and PDF page number so that a response can be traced back to its evidence.

Grounded answers

A grounded answer should be supported by the retrieved passages. A source reference helps the reader verify the answer, but the presence of a citation alone does not guarantee that the citation supports every statement.

When evidence is insufficient

If the document does not contain evidence for a question, the tool should abstain instead of inventing a response. This project uses the exact fallback message: "I could not find this in the document". A similarity threshold is a useful heuristic, and its limitations should be documented.

Evaluation

A retrieval evaluation set records a question, the expected supporting page or answer, the retrieved passages, and whether the passages contain the expected evidence. Comparing chunk sizes helps show the trade-off between focused retrieval and retained context.